Delivery to a target

How wheel turns carry our delivery 40 cm

Recap from last session

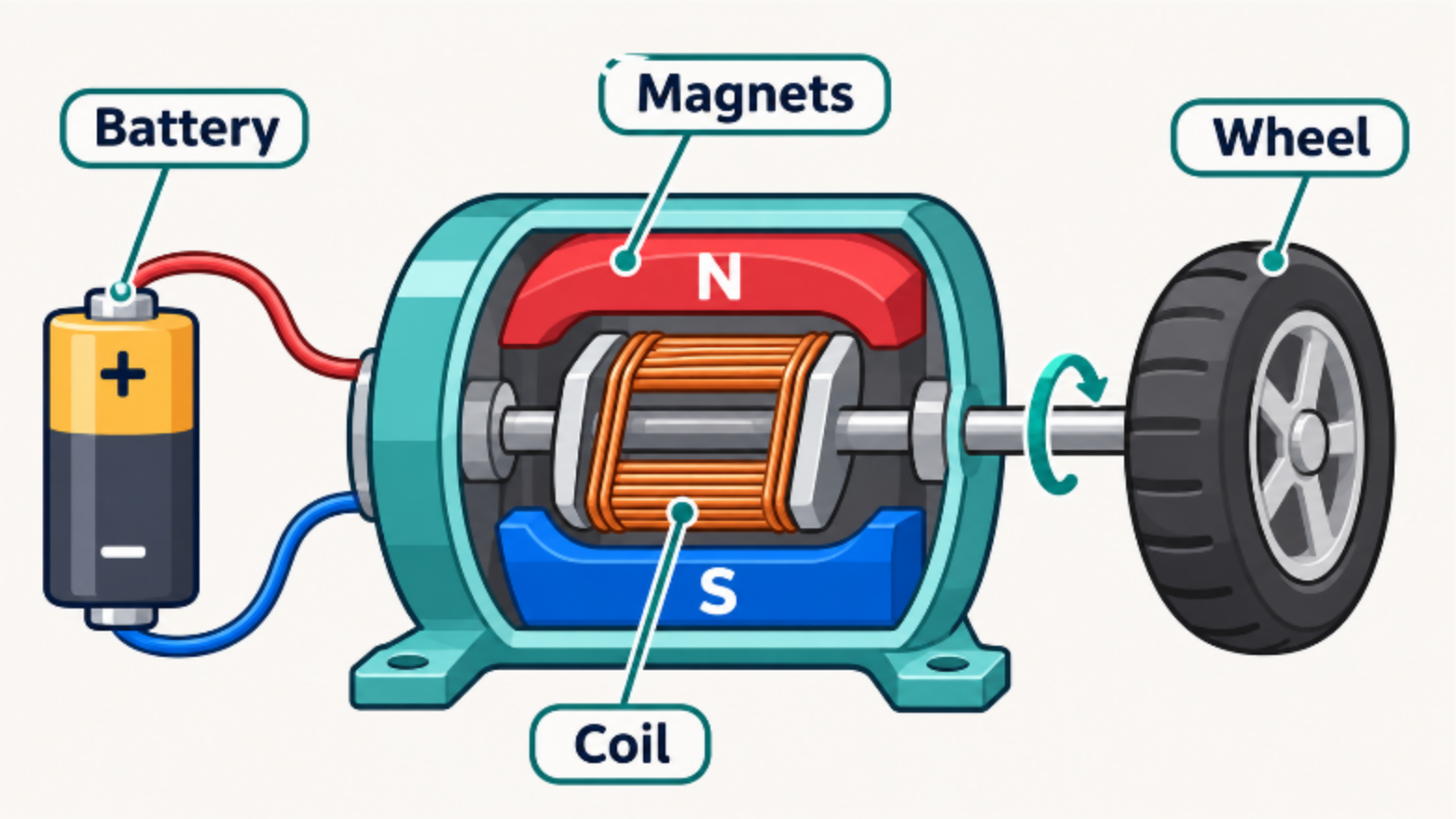
What did our recipe make the robot do?

```
await signal("Y")
await pause(500)
await drive(90)
```

Point to the hub, a motor and its wheel.

Today: connect wheel turns to a measured distance.

What makes the robot move



Electricity flows through the coil.

Magnetic forces turn the motor shaft and the wheel.

The tire grips the floor and moves the robot forward.

Simplified motor picture. The hub controls the motors on our robot.

Degrees measure a wheel turn



90°

Quarter turn



180°

Half turn

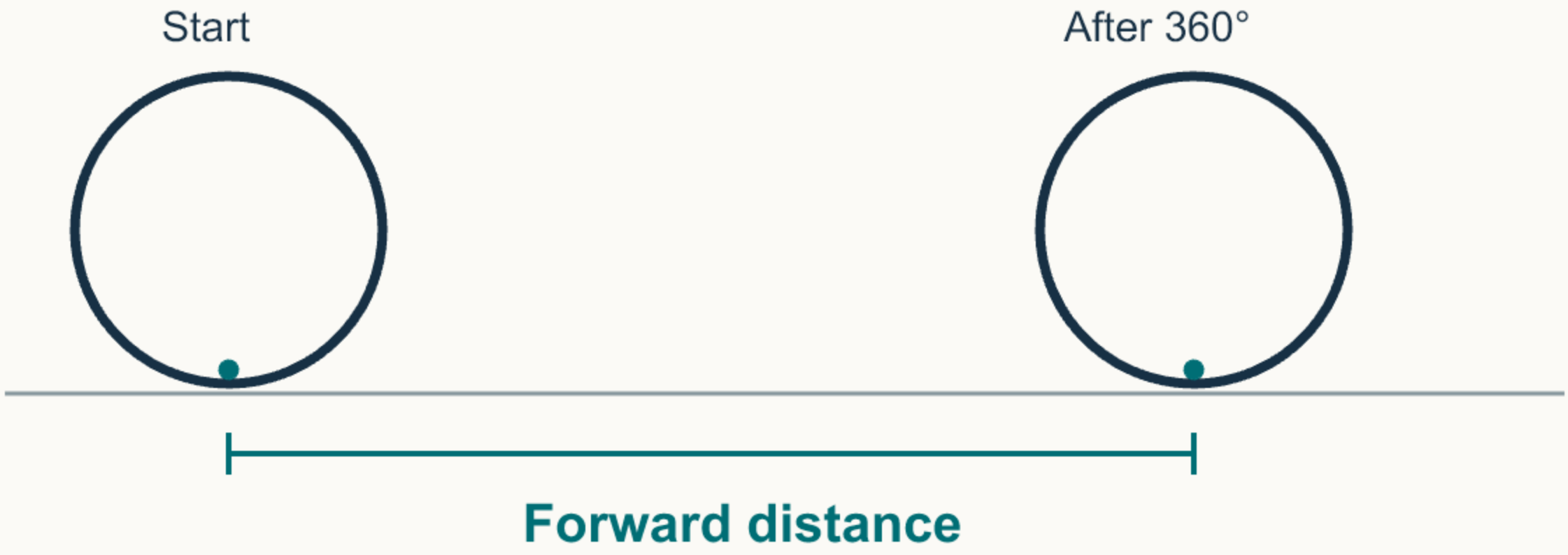


360°

Full turn

Wheel degrees describe how much the wheels rotate.

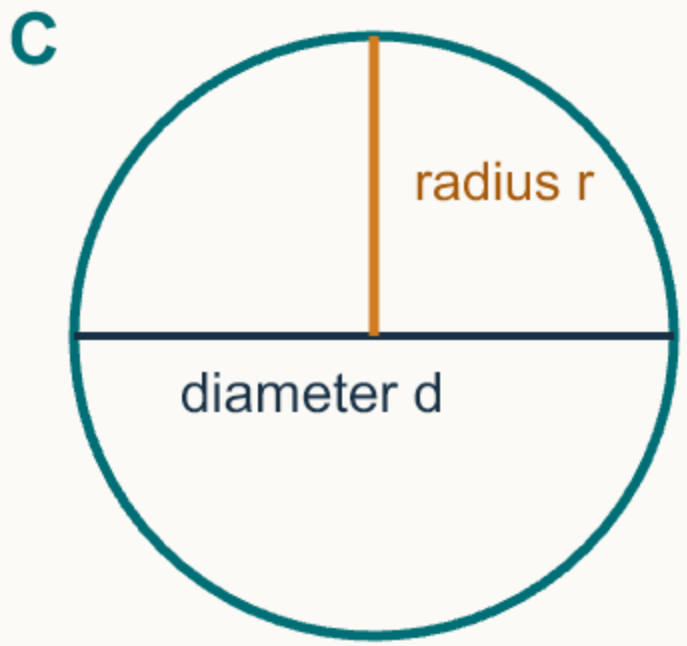
One rotation of the wheel



The dot returns to the same place after one full turn.

Prediction for straight rolling without slipping.

Circumference is the distance around



$$C = \pi \times d$$

$$d = 2 \times r$$

π is about 3.14

Example: d = 5.6 cm
C ≈ 3.14 × 5.6 ≈ 17.6 cm

Measure the actual tire on each kit. Your wheel size may differ.

Wheel degrees to distance

Distance = (wheel degrees ÷ 360) × C

Example circumference C = 17.6 cm

Wheel degrees	Turns	Predicted distance
360°	1	17.6 cm
720°	2	35.2 cm

For YOUR kit, how many centimeters should 180° travel?

Target distance to wheel degrees

Wheel degrees = (target distance ÷ C) × 360

Example: target 40 cm, circumference 17.6 cm

40 cm ÷ 17.6 cm ≈ 2.27 rotations

(40 ÷ 17.6) × 360 ≈ 818 wheel degrees

Use our kit’s C (circumference). Round only the final answer.

A meter is 100 cm. Keep target distance and C in the same unit.

Tell the code how far the wheels turn

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Distance = (wheel degrees ÷ 360) × C

Our prediction

Using our C (circumference), predict cm for 180° and 360°.

What wheel_degrees value should aim for 40 cm?

Recipe fragment: load the complete supplied Week 2 program

Our robot and build jobs

Driver edits and starts after “clear.”

Navigator predicts and resets the stopped robot.

Observer watches, measures and records.

Other team builds. Everyone rotates jobs.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Programming exercise: reach 40 cm

1 Predict and test 180 wheel degrees.

2 Predict and test 360. Swap the driver.

3 Calculate and enter a code value for the 40 cm target.

Wheel degrees = (40 ÷ C) × 360

4 Record predicted cm and measured cm. Explain a change.

Use your kit’s C. Keep speed, load and start mark the same.

Build exercise: a cart that moves freely

The cart carries the cube without hindering robot movement.

Build a flat support with a low edge and a secure mount.

Check: wheels turn freely; nothing drags or snags cables.

Tilt the loose cart gently. Improve one part of the build.

Mentor checks the fit on the stopped robot; then test a short run.

Rotate builder, designer and tester. Keep the base assembled.

Compare prediction with measurement

We set wheel_degrees to _____.

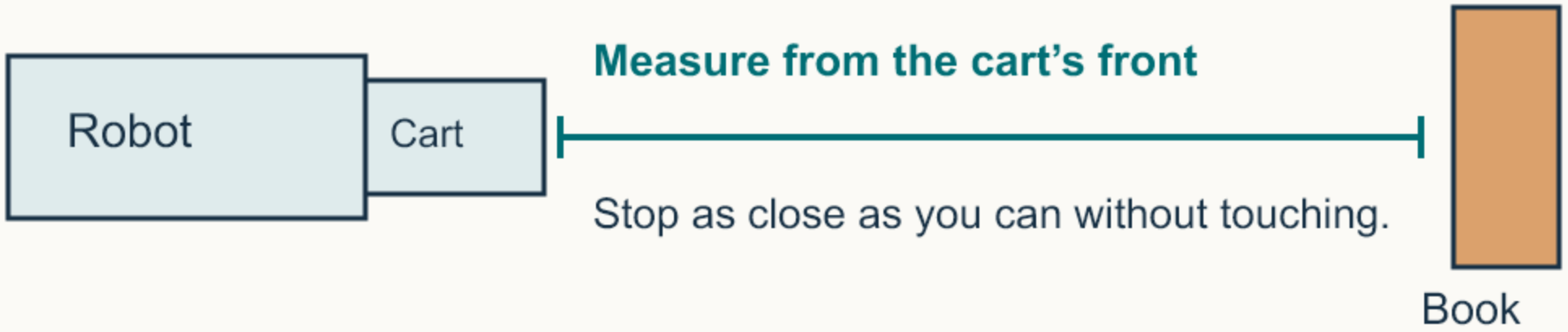
We predicted _____ cm and measured _____ cm.

Our 40 cm run was _____ cm too short / too far.

What one change should we test next?

Optional team challenge: park by a book

Builders attach the cart. The instructor chooses a new start and book position.



Team: work out how wheel_degrees must change.

Programmer: edit, predict and test. Measure the gap left.

Mentor-supervised table lane, well inside all edges. Core work first.

Explain the motion

What makes the motor turn?

How far should half a wheel turn go?

How would you choose the code number for a target?

Save your team project. Store your build. Reset the robot and parts.